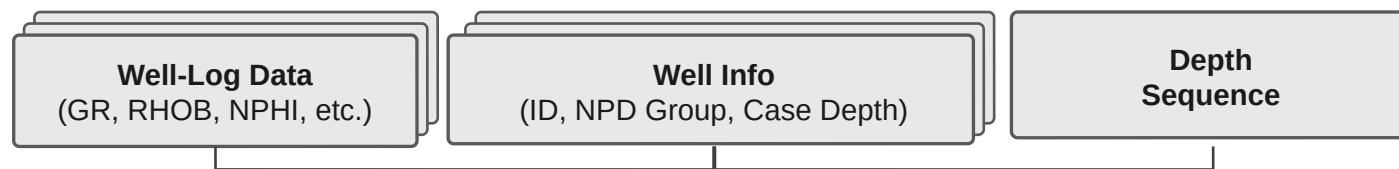


GeoMERIT: A Four-Layer Robust Prediction Framework



Missingness-Aware Geologic State Representation ($\phi(x_i)$)

1

Missingness and outlier detection
($|z| > 5$)

Sentinel Encoding
([$-999, -777$, Indicators])

Well-wise Robust Normalization
(median/MAD)

Multi-scale Depth Context
(Window Stats, Gradient $\nabla c(i)$)

Curve-suite fingerprint

Data Challenge:
non-random curve absence and inter-well heterogeneity

$\phi(x_i)$

Risk-Weighted Learning ($w(i)$)

2

Sample Weight Construction

$$w(i) = w_{class}(y(i)) \cdot w_{bnd}(i) \cdot w_{conf}(i) \quad \left\{ \begin{array}{l} \text{Class-balanced (Effective Numbers)} \\ \text{Boundary-enhanced (Boundary 1.2, Interior 0.7)} \\ \text{Confidence-aware (Based on interpreting confidence 1/2/3)} \end{array} \right.$$

Data Challenge:
Long-tailed labels
ambiguous boundaries

Weighted Data

Structured Posterior Modeling (Ensemble, Hierarchy & Tail Experts)

3

Three-Backend Ensemble
(LightGBM, XGBoost, CatBoost)

Coarse-to-Fine Hierarchy
(6 Super-classes $\rightarrow$ 12 Fine-grained Classes)

Geometric Fusion (Product of Experts)

$$\hat{p}_{fusion} \propto \exp\left(\sum a_m \log p^{(m)}\right)$$

One-vs-Rest Tail Experts

$$\tilde{y}(y = k|\phi) = \hat{p}(y = k|\phi)(1 + \gamma_k 1[s_k > \theta_k])$$

Data Challenge:
Rare-class suppression and adjacent-class ambiguity

$\beta(y = k|\phi(x_i))$

Geological Cost-Guided Decision ($\hat{y}_i$)

4

Logit Adjustment
(Prior Shift Calibration, temperature τ)
 $\hat{p}_{LA}(y = k|\phi) \propto \hat{p}(y = k|\phi) \cdot \pi_k^{-\tau}$

Bayesian Risk Decoding (Expected Cost Minimization)

$$\hat{y}_i = \arg \min_{j \in \{1, \dots, 12\}} \sum_{k=1}^{12} A_{kj} \hat{p}(y = k|\phi(x_i))$$

Data Challenge:
Asymmetric geological misclassification cost

Robust Prediction $\hat{y}_i$